

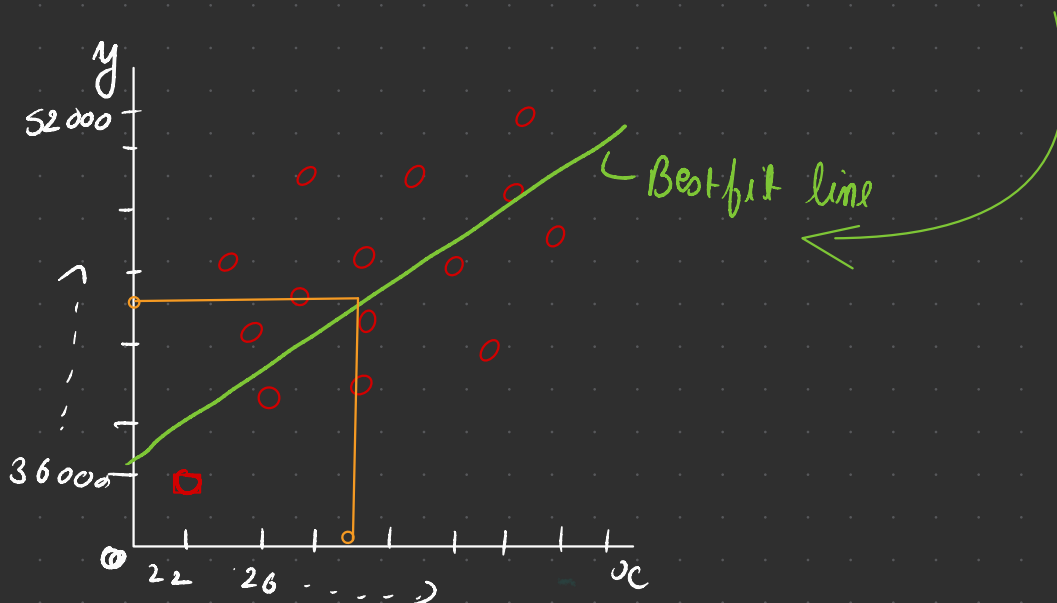
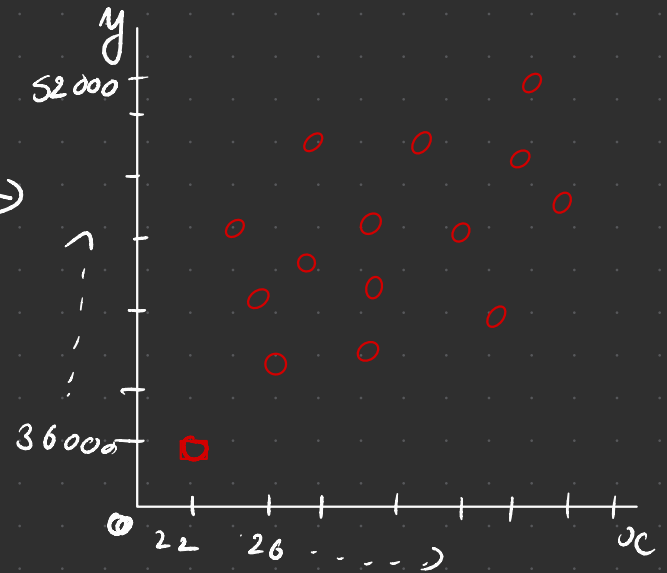
Linear Regression

Problem statement

input ←

Age	Salary
22	36000
26	38000
28	40000
34	48000
39	52000
42	⋮
46	⋮

→ output



Now lets see how you actually create a best fit line

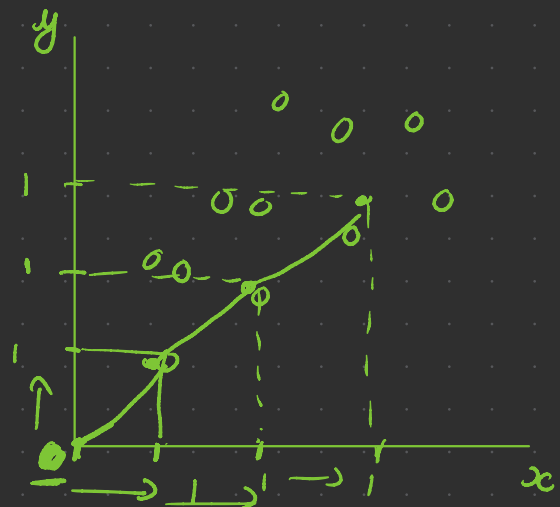
$$y = mx + b$$

m = slope

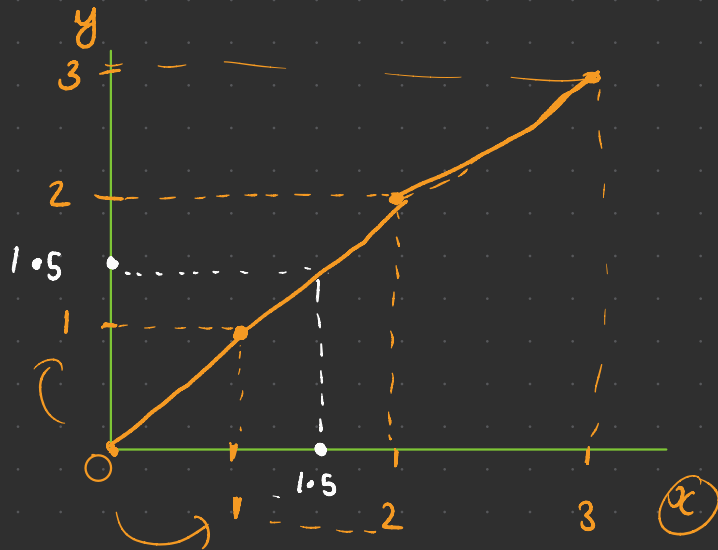
b = intercept

x ⇒ Data point

y ⇒ Prediction



slope $m = 1$
intercept $b = 0$

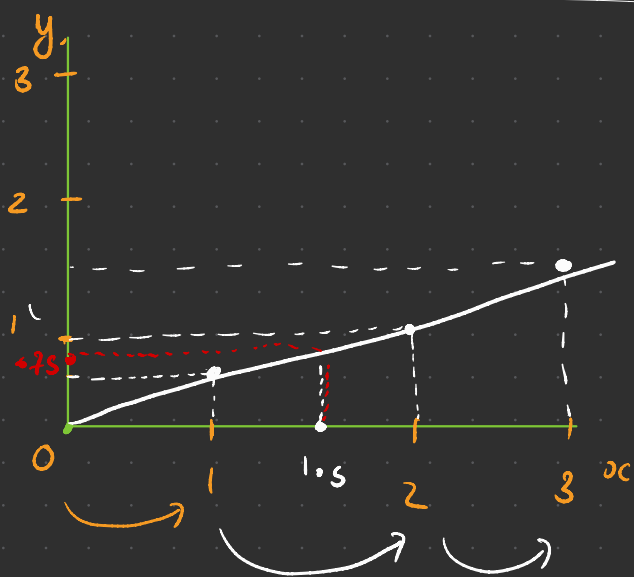


$y = x = 1.5$

$$y = mx + b$$

$$y = 1(1.5) + 0 = y = 1.5$$

$m = 0.5$
 $b = 0$



$x = 1.5 = ? y$

$$y = mx + b$$

$$y = 0.5(1.5) + 0$$

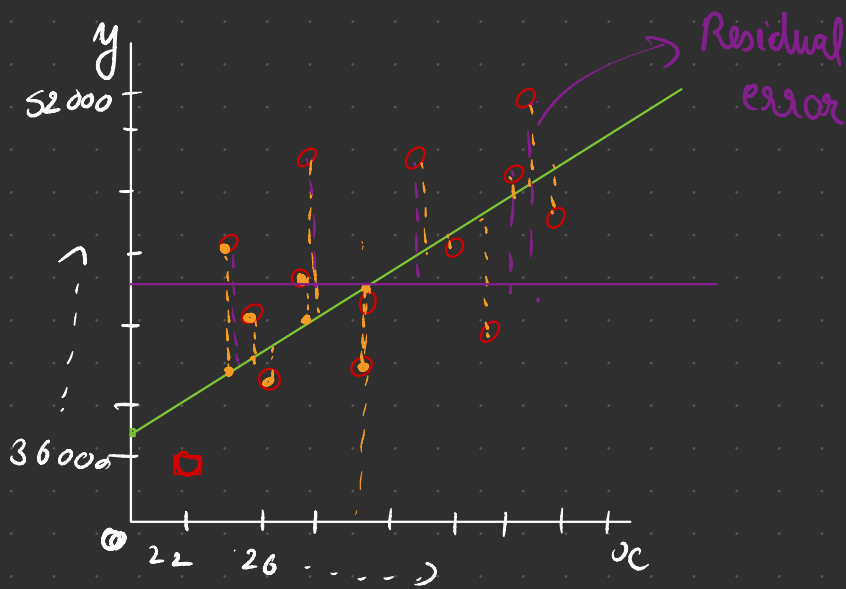
$$y = 0.75$$

$$h(\underline{x}) = \theta_0 + \theta_1 x$$

$\theta_0 \Rightarrow$ intercept

$\theta_1 \Rightarrow$ slope

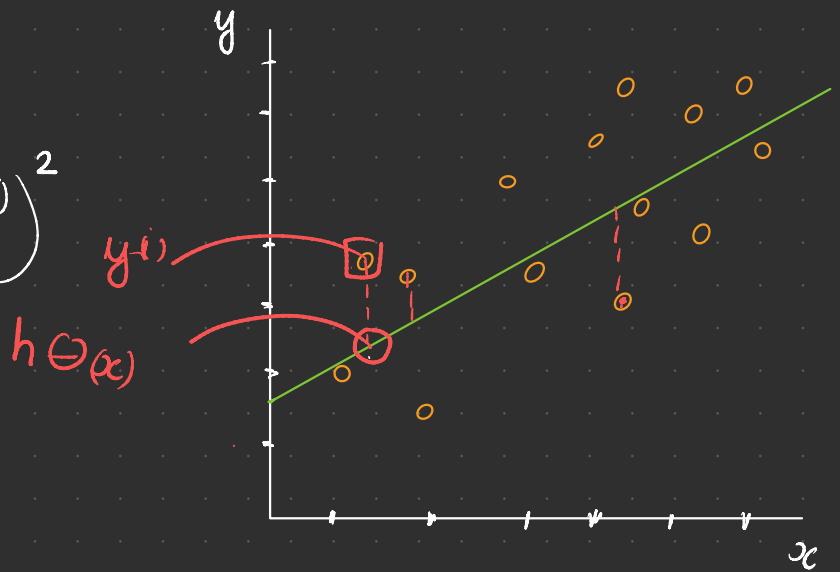
2) have to find the perfect value of θ_0 and θ_1 .



Cost function

$$J(\theta_0, \theta_1) = \sum_{i=1}^m \frac{1}{2m} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Cost function



You have to minimize the cost function by changing the value of θ_0, θ_1 .

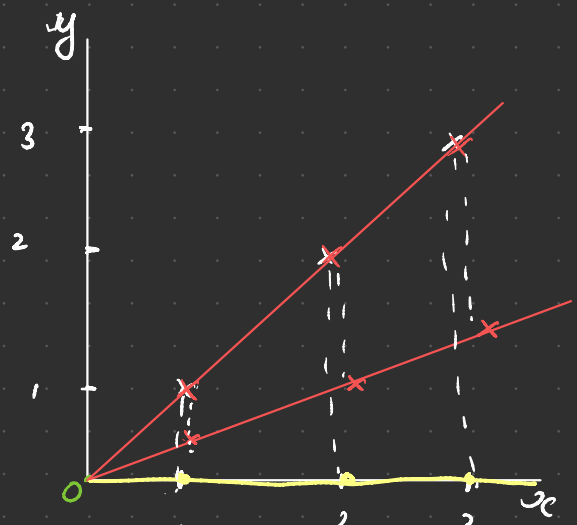
Gradient Descent

$$\begin{aligned} & \begin{matrix} x & y \\ \{1, 1\} \\ \{2, 2\} \\ \{3, 3\} \end{matrix} \end{aligned}$$

$$\theta_0 = 0, \theta_1 = 1$$

$$(h_{\theta}) = \theta_0 + \theta_1 x$$

$$h_{\theta} = 0 + 1(1) = 1$$



$$h(\theta) = 0 + 1(2) = 2$$

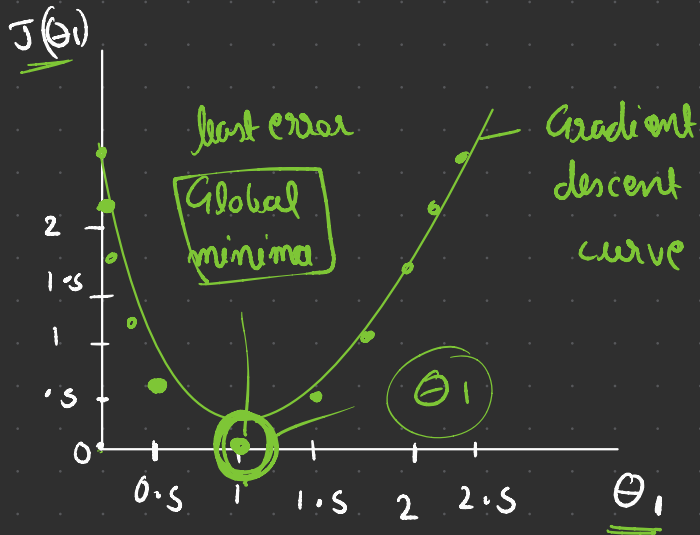
$$h(\theta) = 0 + 1(3) = 3$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h(x^{(i)}) - y^{(i)})^2$$

$$= \frac{1}{6} [(1-1)^2 + (2-2)^2 + (3-3)^2]$$

$$= \frac{1}{6} (0^2 + 0^2 + 0^2)$$

$$= 0 \quad J(\theta_0) = 0$$



$$\theta_1 = 0.5, \theta_0 = 0$$

$$J(\theta) = \frac{1}{6} [(1-0.5)^2 + (1-2)^2 + (3-1.5)^2] =$$

$$0 + 0.5(1) = 0.5$$

$$0 + 0.5(2) = 1$$

$$0 + 0.5(3) = 1.5$$

$$\approx 0.58$$

$$\theta_1 = 0, \theta_0 = 0$$

$$h(\theta) = \theta_0 + \theta_1 x_1$$

$$0 + 0 = 0$$

$$0 + 0 = 0$$

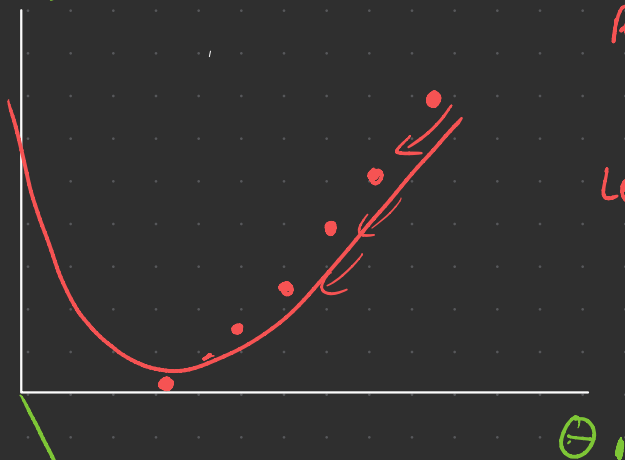
$$0 + 0 = 0$$

$$J(\theta_1) = \frac{1}{6} [(0-1)^2 + (0-2)^2 + (0-3)^2]$$

$$J(\theta_1) = 3.2 - 3$$

$$J(\theta_1, \theta_0)$$

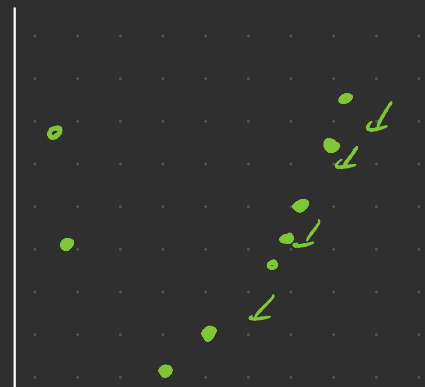
$J(\theta_1)$



Repeat convergence
Theorem

Learning Rate

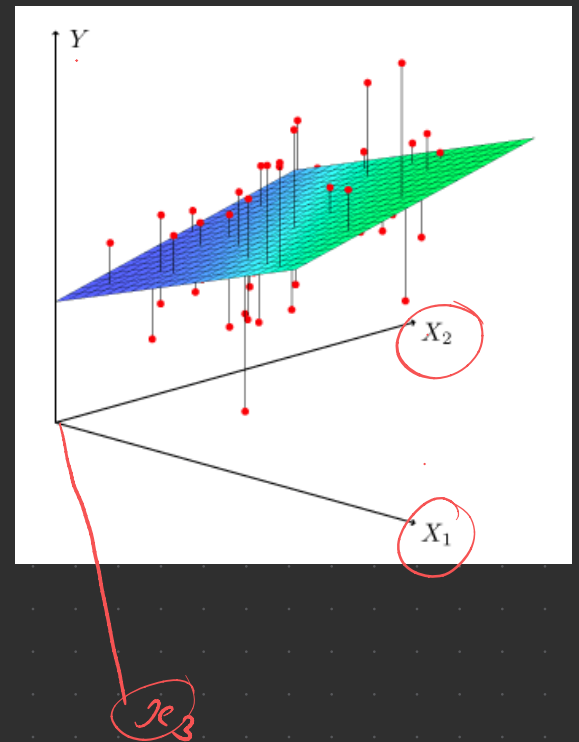
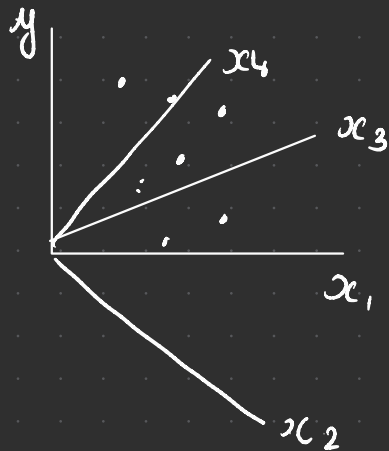
α



$$y = mx + b$$

$$h(\theta) = \theta_0 + \theta_1 x_1$$

$$h(\theta) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$



Performance matrices

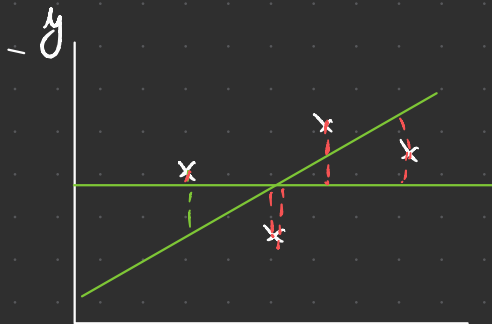
R^2

Adjusted R^2

$$R^2 = 1 - \frac{\text{Sum of Residuals}}{\text{Sum of total}} \Rightarrow 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$h(\theta)$

$\uparrow$
 $\sum (y_i - \hat{y}_i)^2$
 $\sum (y_i - \bar{y})^2$



The Result will be always positive
The result will be in Decimal

$$0.86 = 86\%$$

Location Age $\rightarrow$ salary $\rightarrow$ sleep

$$R^2 = 81\%$$

$$R^2 = 88\%$$

$$R^2 = 89\% \leftarrow \text{Model}$$

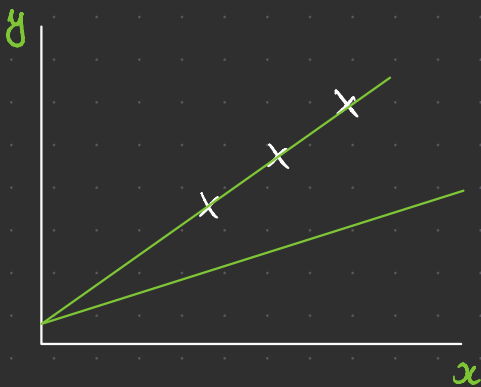
Adjusted R^2

$$\Rightarrow \frac{1 - (1 - R^2)(n - 1)}{N - \underline{P} - 1}$$

$n = \text{no. of Rows}$
 $P = \text{features}$



R^2	AR^2	
81%	79%	$P=1$
88%	89%	$P=2$
↑↑ 89%	86% ↓↓	



$$J(\theta_0, \theta_1) = \underline{0}$$

overfitting

Training → model Perform better
 Testing → model Not Performed Good
 (low bias)
 (high variance)

underfitting

Training → model Perform is bad
 Testing → model Not Performed Good
 (high bias)
 (high variance)

M_1

Training → 92%
 Testing → 64%

↓

Overfitting

M_2

Training → 92%
 Testing → 90%

↓

Perfect model
 (low bias)
 (low variance)

↓↓

M_3

Training → 68%
 Testing → 64%

↓

underfitting

Ridge and Lasso Regression

cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left[(h_{\theta} x^{(i)}) - y^{(i)} \right]^2 + \lambda (\text{slope})^2$$

~~= 0~~ $\rightarrow$

